BEU CONNECT

B.Tech 6th Semester Exam., 2022

(New Course)

Time: 3 Hours

Code: 105602 *Subject: Machine Learning* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer/Answer the following (Any seven): **[2×7=14]**

(a) Which of the following is false regarding regression?

(i) It relates inputs to outputs.
(ii) It is used for prediction.
(iii) It may be used for interpretation.
(iv) It discovers causal relationships.

(b) Consider a linear-regression model with N = 3 and D = 1 with input-output pairs as follows: y1 = 22, x1 = 1, y2 = 3, x2 = 1, y3 = 3, x3 = 2. What is the gradient of mean-square error (MSE) with respect to β1 when β0 = 0 and β1 = 1? Give your answer correct to two decimal digits.

(c) You observe the following while fitting a linear regression to the data: As you increase the amount of training data, the test error decreases and the training error increases. The train error is quite low (almost what you expect it to), while the test error is much higher than the train error. What do you think is the main reason behind this behaviour? Choose the most probable option:

(i) High variance
(ii) High model bias
(iii) High estimation bias
(iv) None of the above

(d) The square of the correlation coefficient r^2 will always be positive and is called the

(i) regression
(ii) coefficient of determination
(iii) KNN
(iv) algorithm

(e) The parameter β0 is termed as intercept term and the parameter β1 is termed as slope parameter. These parameters are usually called as

(i) regressionists
(ii) coefficients
(iii) regressive
(iv) regression coefficients

(f) In order to calculate confidence intervals and hypothesis tests, it is assumed that the errors are independent and normally distributed with mean zero and

(i) mean
(ii) variance
(iii) SD
(iv) KNN

(g) Which of the following is true about residuals?

(i) Lower is better
(ii) Higher is better
(iii) (i) or (ii) depending on the situation
(iv) None of the above

(h) Suppose that we have N independent variables (X1, X2, ..., Xn) and dependent variable is Y. Now imagine that you are applying linear regression by fitting the best fit line using least square error on this data. You found that correlation coefficient for one of its variables (Say X1) with Y is -0.95. Which of the following is true for X1?

(i) Relation between X1 and Y is weak
(ii) Relation between X1 and Y is strong
(iii) Relation between X1 and Y is neutral
(iv) Correlation cannot judge the relationship

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(i) Suppose you plotted a scatter plot between the residuals and predicted values in linear regression and you found that there is a relationship between them. Which of the following conclusions would you make about this situation?

(i) Since there is a relationship, it means our model is not good
(ii) Since there is a relationship, it means our model is good
(iii) Cannot say
(iv) None of the above

(j) What would be the root mean square training error for this data if you run a linear regression model of the form (Y = A0 + A1X)?

(i) Less than zero
(ii) Greater than zero
(iii) Equal to zero
(iv) None of the above

Q.2 (a) Discuss various challenges in machine learning based applications with suitable examples. **[7]**

(b) Consider a learning problem where each instance is described by a conjunction of n Boolean attributes a1 ⋯ an. Thus, a typical instance would be (a1 = T) ∧ (a2 = F) ∧ ⋯ ∧ (an = T). Now consider a hypothesis space H in which each hypothesis is a disjunction of constraints over these attributes. For example, a typical hypothesis would be (a1 = T) ∨ (a5 = F) ∨ (a7 = T). Propose an algorithm that accepts a sequence of training examples and outputs a consistent hypothesis if one exists. Your algorithm should run in time that is polynomial in n and in the number of training examples. **[7]**

Q.3 (a) Consider the following set of training examples: **[6]**

Instance	Classification	a1	a2
1	+	T	T
2	+	T	T
3	−	T	F
4	+	F	F
5	−	F	T
6	−	F	T

(a) What is the entropy of this collection of training examples with respect to the target function classification?

(b) What is the information gain of a2 relative to these training examples?

(b) Design a two-input perceptron that implements the Boolean function A ∧ B. Design a two-layer network of perceptrons that implements A XOR B. **[8]**

Q.4 (a) Consider a two-layer feedforward ANN with two inputs a and b, one hidden unit c, and one output unit d. This network has five weights (wca, wcb, wc0, wdc, wd0), where wx0 represents the threshold weight for unit x. Initialize these weights to the values (·.1, ·.1, ·.1, ·.1, ·.1), then give their values after each of the first two training iterations of the BACKPROPAGATION algorithm. Assume learning rate η = 0.3, momentum α = 0.9, incremental weight updates, and the following training examples: **[7]**

a	b	d
1	0	1
0	1	0

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(b) Draw two-class, two-dimensional data such that (i) PCA and LDA find the same direction and (ii) PCA and LDA find totally different directions. **[7]**

Q.5 (a) Suppose you test a hypothesis h and find that it commits r = 300 errors on a sample S of n = 1000 randomly drawn test examples. What is the standard deviation in errors(h)? How does this compare to the standard deviation in the example at the end of Section 5.3.4? **[6]**

(b) Considering the training data provided in the following table, try to build an associative classifier model by generating all relevant association rules with support and confidence thresholds 10% and 60% respectively. Classify using this model the new example: age ≤ 30, income = medium, student = yes, credit-rating = fair, selecting the rule with the highest confidence. What would be the classification if we choose to vote the class among all rules that apply? **[8]**

RID	age	income	student	credit_rating	class:buys_computer
1	≤ 30	high	no	fair	no
2	≤ 30	high	no	excellent	no
3	31...40	high	no	fair	yes
4	> 40	medium	no	fair	yes
5	> 40	low	yes	fair	yes
6	> 40	low	yes	excellent	no
7	31...40	low	yes	excellent	yes
8	≤ 30	medium	no	fair	no
9	≤ 30	low	yes	fair	yes
10	> 40	medium	yes	fair	yes
11	≤ 30	medium	yes	excellent	yes
12	31...40	medium	no	excellent	yes
13	31...40	high	yes	fair	yes
14	> 40	medium	no	excellent	no

Q.6 (a) Draw the Bayesian belief network that represents the conditional independence assumptions of the naive Bayes classifier for the Play Tennis problem stated in the following table. Give the conditional probability table associated with the node wind. **[8]**

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

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(b) What is kernel regression and noise variance? Explain in detail. **[6]**

Q.7 (a) What is the lazy version of the eager decision tree learning algorithm? What are the advantages and disadvantages of your lazy algorithm compare to the original eager algorithm? **[6]**

(b) What is clustering? Why do we use this in machine learning? Discuss the following algorithms: **[8]**

(i) K-means

(ii) K-medoid

Q.8 (a) How can machine learning techniques help to identify forged data recognition application? Explain the method in detail. **[7]**

(b) Differentiate between supervised and unsupervised training. Explain with suitable examples. **[7]**

Q.9 Write short notes on the following: **[4+5+5]**

(i) Linear Algebra for ML

(ii) ANN

(iii) Reinforcement Learning

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